Course structure of B.Tech (Civil Engineering) for batch 2024-28

Semester 5	OE		Open Elective 2 **((NPTEL/SWAYAM/MOOC))**	3	0	0	3	3
	HSC		Engineering Economics	3	0	0	3	3
	PE		Program Elective 1	3	0	0	3	3
	PC		Hydrology and Water Resources	3	0	2	5	4
	PC		Foundation and Geotechnical Applications	3	0	2	5	4
	PC		Design of RCC Structures	3	0	0	3	3
	PC		Computer Aided Design Drawing Lab - I	0	0	2	2	1
				18	**0**	**6**	**24**	**21**
Semester 6	OE		Open Elective 3 (From Other Department of FoET)	3	0	0	3	3
	PE		Program Elective 2	3	0	0	3	3
	PE		Program Elective 3	3	0	0	3	3
	PC		Design of Steel Structures	3	0	0	3	3
	PC		Computer Aided Design Drawing Lab - II	0	0	2	2	1
	PC		Environmental Engineering	3	0	2	5	4
	PC		Highway and Traffic Engineering	3	0	2	5	4
				18	**0**	**6**	**24**	**21**

Course structure of B.Tech (Civil Engineering) for batch 2024-28

Semester 7	Pro		Summer Internship	0	0	0	0	2
	OE		Open Elective 4 (From Other Department of FoET)	3	0	0	3	3
	PE		Program Elective 4	3	0	0	3	3
	PE		Program Elective 5	3	0	0	3	3
	PC		Estimation Costing Contracts and Valuations	3	1	0	4	4

	PC		Project Management	3	0	2	5	4
	PC		Earthquake Engineering	3	1	0	4	4
	Pro		**Seminar**					1
				18	**2**	**2**		**24**
Semester 8	**Pro**		Major/Comprehensive Project					**12**

					Estimation Costing and Valuation					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs./Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	1	0	4	4	25	50	25	--	--	100

COURSE OBJECTIVES

1. To learn the concepts of Rate-Analysis.
2. To learn the concepts of Specification and its important.
3. To learn the fundamentals of Estimation.
4. To learn the concepts of Costing and its Description Writing.

UNIT I: RATE ANALYSIS **16 Hrs.**

Introduction, Importance of Rate Analysis, Calculation of Rate Analysis, Schedules of Rate, Task Work, Rate Analysis of Different Items – Excavation, Brick Masonry, Stone Masonry, Cement Concrete (C.C) Work, R.C.C. Work, Plastering Work, Flooring Work, White Washing and Colouring Work, Roofing Work, Other Miscellaneous Items.

UNIT II: SPECIFICATION **12 Hrs.**

Introduction, Objectives of Specification, Importance of Specification, Types of Specification, Principles of Writing Specification, Standard Specification Writing for Construction Items – Earthwork for Foundation, Cement Concrete, R.C.C Work, Brick Work in Cement Mortar (Class I, II and III), Different Types of Flooring, Different Types of Stone Masonry, Colouring Items, Roofing and Terrace Items (A.C. Sheet, G.I. Sheets), Canal Earthwork, etc.

UNIT III: ESTIMATION OF BUILDING AND OTHER WORK **16 Hrs.**

Introduction, Types of Estimate, Methods of Estimation, General Estimation and Costing of Residential Building (Single Story and Double Story), Estimation of Septic Tank, Culverts, R.C.C Structures (Slab, Beam, Column-Footing, Retaining Wall), Road Work (Quantity of Cutting and Filling).

UNIT IV: VALUATION CONTRACT AND ARBITRATIONS **12 Hrs.**

Definitions of various terms- method of valuation- Freehold & Leasehold properties- Sinking fund- depreciation and method of estimating depreciation- Outgoings, types of contract and arbitrations.

TOTAL HOURS: 56 Hrs.

COURSE OUTCOMES

On completion of the course, student will be able to:

CO-1: Remember the Fundamentals Concepts of Estimation.
CO-2: Demonstrate the Aspects of Rate Analysis, Specification and Estimation.
CO-3: Utilise the Concepts to Identify the Rates of Different Construction Items.
CO-4: Analyse the Different Construction Items and Write Its Specification.
CO-5: Evaluate the Building and Its Components Using Concepts of Estimation.
CO-6: Evaluate and Estimate Different Construction.

TEXT-BOOK AND REFERENCE BOOKS:

1. B. N. Dutta, "**Estimating and Costing in Civil Engineering**:, 28th Revised Edition, New Delhi: CBS Publishers & Distributors Pvt. Ltd.
2. M. Chakraborty, "**Estimating, Costing, Specification & Valuation in Civil Engineering,**" Kolkata.
3. D. D. Kohli, and R. C. Kohli, "**A Text Book of Estimating and Costing (Civil)**", S Chand Publishers.
4. S.C. Rangwala, "**Estimating, Costing and Valuation, 15th Edition**", Charotar Publishing House Pvt. Ltd.
5. Peurifoy, R.L., Ledbetter, W.B. and Schexnayder, C., "**Construction Planning, Equipment and Methods"**, 6th Edition, Tata McGraw-Hill, New Delhi, 2003
6. C.P.W.D. Hand Books
7. IS: 1200 (Part 1 to 28), "**Methods of Measurement of Building and Civil Engineering Works**", Bureau of Indian Standards
8. SOR of respective Financial Year as approved by Authority

Teaching Scheme					Project Management Examination Scheme					
L	T	P	C	Hrs./Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	2	4	5	25	50	25	--	--	100

COURSE OBJECTIVES

- To develop ability and knowledge about the philosophy, concepts and scope of project management and the phases of project life cycle.
- To develop ability and knowledge about the different forms of project organization structures and work breakdown structures.
- To create problem solving ability and develop knowledge about the project planning & scheduling tools and also about the basic project controlling methods
- To develop ability and knowledge about the methods for project monitoring and methodology for project risk management.

UNIT 1: CONCEPT, SCOPE OF PROJECT MANAGEMENT AND PHASES OF PROJECT LIFE CYCLE **12 Hrs.**

Introduction- Parameters affecting a project- Project planning & implementation cycle- Concept & scope of project management-Role of project manager- Enhancing the probability of success of a project Phases of project life cycle: Idea, Feasibility - Development, Implementation and Operation. Work Break Down structure (WBS)- Role of project manager in developing WBS- Typical hierarchy in the WBS of a project- Product oriented WBS; Functionally oriented WBS.

UNIT 2: PROJECT PLANNING AND SCHEDULING **12 Hrs.**

Project organization structures - Factors responsible for organizational revolution- Formal & informal organization structures- Requirements of a project organization- Matrix organization structure- Selecting a project organization structure- Criteria to help determine a suitable organizational form in a given project environment Project Planning & Scheduling: Scheduling principles- Bar charts (Gantt charts)- Milestones Charts- S-curve- Network logic diagram- Critical path method- Arrow diagram- Time Estimates- Slack- Total float- Free float - Independent floats- Case studies.

UNIT 3: PROBABILISTIC TOOLS FOR PROJECT SCHEDULING AND PROJECT CONTROL **16 Hrs.**

Probabilistic tools for Project Scheduling - PERT (Project Evaluation & Review Techniques) - Three time estimates (optimistic, most likely, pessimistic)- Beta distribution- Expected time- Variance in project duration- Case Studies Project Control: Concept- Control cycle- Basic controlling parameters- Line of Balance- Role of project management on control cycle- Time control-Cost control-Potentiality of cost reduction during different phases of a project- Cost planning- Control curves- Cash flow- Network Crashing, Time cost trade-off planning for minimum costs- Cost slope concept- Crash point- Normal Point- Total project cost- Controlling cost overrun & time overrun.

UNIT 4: PROJECT MONITORING AND RISK MANAGEMENT **16 Hrs.**

Project Monitoring: Measurement of performance- Reporting of performance- Corrective measures for in favorable variations- Major functions of monitoring- Influence of decision-making authority in project monitoring Project Risk Management: Risk identification- Risk analysis- Risk response planning and mitigation measures- Case studies Sustainable development: Project management for sustainable development

Max. 56 Hrs

COURSE OUTCOMES

CO-1: Demonstrate the concepts and philosophy of project management and also about the different phases of the project life cycle.
CO-2: Design the methodology for formulation and application of work breakdown structure and organization structure.
CO-3: Demonstrate project planning and scheduling tools and techniques and complex critical path network diagrams.
CO-4: Analyze the problems pertaining to project evaluation and review technique.
CO-5: Apply the project monitoring methods and tools like Earned Value Analysis.
CO-6: Develop concepts about project controlling methods, project risk management and project management for sustainable development.

TEXT-BOOK AND REFERENCE BOOKS:

1. Kumar Neeraj Jha, "**Construction Project Management**", Pearson Publishers, New Delhi, 2018.
2. Nicholas John M, "**Project Management for Business and Technology: Principles and Practice**", 2nd Edition, Pearson Prentice Hall New Delhi, 2007.
3. Iyer P Parameshwar, "**Engineering Project Management with case studies**". Wheeler Publishing New Delhi, 2001.
4. Joy PK, "**Handbook of Construction Management**", Macmillan Delhi, 1990

					Project Management Laboratory					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs./Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	2	4	5	25	50	25	--	--	100

COURSE OBJECTIVES

1. Evaluate project management software like Microsoft Project, Primavera P6, etc., and learn their features.
2. Develop work breakdown structures (WBS) in Microsoft Project, assign resources, and use scheduling techniques.
3. Track project progress, analyze performance using EVM, and generate reports in Microsoft Project.
4. Explore best practices in using Microsoft Project for construction management, including visualization, managing multiple projects, and reporting, and learn about the Monte Carlo process for risk analysis.

UNIT 1: INTRODUCTION TO PROJECT MANAGEMENT TOOLS **4 Hrs.**

Introduction- Overview of project management software, including Microsoft Project, Primavera P6 etc; Comparison of different project management software options; Introduction to Microsoft Project interface and features; Creating a new project; Setting up project properties and options; Adding tasks, task durations, and dependencies; Creating a project timeline using Gantt charts.

UNIT 2: PROJECT PLANNING AND SCHEDULING **8 Hrs.**

Developing a work breakdown structure (WBS) in Microsoft Project; Assigning resources to tasks; Understanding resource calendars and resource leveling; Setting task constraints and deadlines; Fine-tuning project schedules using task types and effort-driven scheduling; Analyzing project schedules using the critical path method (CPM) and resource histograms.

UNIT 3: PROJECT TRACKING AND MONITORING **8 Hrs.**

Updating project progress and tracking task completion; Managing changes to project scope, schedule, and resources, Analyzing project performance using earned value management (EVM); Generating project reports and dashboards; Communicating project status to stakeholders using Microsoft Project.

UNIT 4: ADVANCED TOPICS IN CONSTRUCTION MANAGEMENT **8 Hrs.**

Best practices for using Microsoft Project in construction project management; Visualization techniques in Microsoft Project for better project understanding; Managing multiple projects with Microsoft Project Server or Project Online; Reporting capabilities of Microsoft Project for effective communication and decision-making. Introduction to Monte Carlo process for project risk analysis.

Sustainable development: Project management for sustainable development

Max. 28 Hrs

COURSE OUTCOMES

CO-1: Evaluate and select appropriate project management software for specific project requirements.
CO-2: Create comprehensive project plans using work breakdown structures and interpret gant charts.
CO-3: Assign Material, Manpower and Machinery Resources and Apply Levelling Techniques.
CO-4: Use Earned Value Management techniques to monitor project progress, analyze variances.
CO-5: Apply construction project management best practices, including visualization and managing multiple projects.
CO-6: Utilize Microsoft Project for effective project communication, decision-making, and risk analysis.

TEXT/REFERENCE BOOKS

1. "**Microsoft Project 2019 Step by Step**" by Cindy Lewis and Carl Chatfield.
2. "**A Guide to the Project Management Body of Knowledge (PMBOK® Guide)–Sixth Edition**" by Project Management Institute.
3. "**Construction Project Management: A Complete Introduction**" by Alison Dykstra.
4. "**Project Management for Engineering, Business and Technology**" by John M. Nicholas and Herman Steyn.

					Earthquake Engineering					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs./Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	1	0	4	4	25	50	25	--	--	100

COURSE OBJECTIVES

1. To understand the fundamentals of earthquake and seismological aspects
2. To understand free and forced vibration and their analysis
3. To explain the principles of earthquake-resistant structures
4. To discuss code provisions and their application to different types of structures

UNIT I: SEISMOLOGY AND EARTHQUAKE FUNDAMENTALS **12 Hrs.**

Seismology and earthquakes**:** Basic earthquake principles: Introduction - Internal structure of earth - Plate tectonics faults - seismic waves – Seismograph - Classification of earthquakes - Magnitude and intensity of earthquakes - Seismic zones in India - Earthquake ground motion: Amplitude parameters - frequency content parameters - duration parameters, Common Earthquake effects: Surface rupture - Regional subsidence – liquefaction – slope movement – Tsunami

UNIT II: THEORY OF VIBRATION AND MACHINE FOUNDATION **16 Hrs.**

Free Vibrations of Single-Degree-of-Freedom Systems: Dynamic loads and dynamic analysis - degrees of freedom - Undamped free vibrations - multiple elastic forces - viscously damped vibrations, equations of motion and solution - logarithmic decrement. Forced vibrations of single-degree-of-freedom systems: Forced vibrations (harmonic loading) of single-degree-of-freedom systems - Undamped and viscously damped vibrations - equations of motion and solution - Force transmitted to the foundation – transmissibility - response to harmonic support excitations. Introduction, Types of Machines and Foundations, General requirements, Permissible Amplitude – Design

UNIT III: EARTHQUAKE-RESISTANT FEATURES OF STRUCTURE **12 Hrs.**

Load Transfer Path - Strength Hierarchy - Reversal of Stresses - Importance of Beam-Column Joints - Importance of Stiffness and Ductility (Capacity Design Concept) in Structures - Earthquake Design Philosophy Inertia Forces - Behaviour of Brick Masonry - Stone Masonry and RC structures - Base Isolation - Adoptive Systems - Case Studies.

Effect of Short Column - Soft Storey - Improper Detailing - Masonry Infill Walls – Eccentricity – Pounding - Floating Columns on Seismic Performance of Building

UNIT IV: LATERAL LOADS ON BUILDINGS AND DUCTILE DETAILING **16 Hrs.**

Analysis for Earthquake Loads: IS: 1893-2016 - Seismic Coefficient method- modal analysis - Applications to multi-storied building frames.

Ductile Detailing: Ductility of R.C. structures- Confinement- detailing as per IS-13920-2016 – principles of design of beams, and columns.

TOTAL HOURS: 56 Hrs.

COURSE OUTCOMES

CO-1: Explain the fundamentals of seismology and earthquake effects
CO-2: Analyse free and forced vibration with applications
CO-3: Design machine foundations
CO-4: Demonstrate the basic principle of earthquake-resistant design
CO-5: Explain the ductile detailing concept and its associated effect on seismic structural response
CO-6: Design the structures subjected to the earthquake by relevant Indian standards.

TEXT-BOOK AND REFERENCE BOOKS:

1. A.K. Chopra, **"Dynamics of structures"**, Prentice Hall.
2. I.S. 1893 - 2016, **"Criteria for Earthquake Resistance Design of Structures"**. BIS
3. Kramer, S. L., **"Geotechnical Earthquake Engineering"**, Pearson Education.
4. Day, R. W., **"Geotechnical Earthquake Engineering Handbook"**, McGraw Hill.
5. Kamlesh Kumar, **"Basic Geotechnical Earthquake Engineering"**, New Age.